Subash Subramani

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Professional Summary

Highly motivated software engineer pursuing an MSc in Artificial Intelligence at Aston University, Birmingham. Experienced in developing AI-powered applications, including Retrieval-Augmented Generation systems and machine learning optimization frameworks. Strong background in full-stack development with growing expertise in LLM integration, semantic search, and scalable AI system design.

Academic Projects

Local Retrieval-Augmented Generation (RAG) Assistant *AI System Development*
Personal Project

- Designed and implemented a Retrieval-Augmented Generation (RAG) system integrating a local LLM with a persistent vector database.
- Built a semantic search pipeline using transformer-based embeddings and document chunking for context-aware question answering.
- Developed a backend API to retrieve relevant knowledge and dynamically augment LLM prompts.
- Implemented persistent vector storage and similarity search for efficient knowledge retrieval.
- Engineered modular architecture supporting local inference and scalable AI integration.
- Technologies: Python, LangChain, Chroma Vector Database, HuggingFace Embeddings, Django REST API.

Budget-Aware Hyperparameter Optimization *MSc Artificial Intelligence*
Aston University, Birmingham, UK

- Designed and implemented a machine learning framework to optimize hyperparameters under strict computational and time budget constraints.
- Applied grid search, random search, and early stopping strategies to balance model performance and resource efficiency.
- Evaluated models using cross-validation and performance metrics to identify optimal configurations.
- Conducted experiments using Python, Scikit-learn, Jupyter Notebook, and Google Colab.
- Reduced training cost while maintaining competitive model accuracy.

Skills

- **Programming:** Python, JavaScript, Node.js
- **Frameworks:** Django, Express.js, React.js
- **AI / Machine Learning:** Retrieval-Augmented Generation (RAG), Semantic Search, Transformer Embeddings, Hyperparameter Optimization, Model Evaluation, Cross-Validation
- **AI Tools:** LangChain, Vector Databases (Chroma), HuggingFace Embeddings
- **Data Science:** NumPy, Pandas, Matplotlib, Scikit-learn
- **Backend Development:** RESTful APIs, LLM Integration, Local Model Deployment
- **Tools:** Git, MySQL, AWS, Jupyter Notebook, Google Colab, Webots
- **Soft Skills:** Problem-solving, Analytical Thinking, Teamwork, Communication
- **Languages:** English (Professional), Tamil (Fluent), Japanese (Beginner)

Work Experience

Software Engineer *Nov 2023 – Aug 2024*
Avelator Solutions India Pvt Ltd, Coimbatore, India

- Developed a Hospital Management System with 8 core modules, including Doctor and Sales Administration.
- Enhanced frontend usability using React.js and optimized MySQL queries, reducing latency by 30%.
- Built RESTful APIs using Node.js and Express.js to enable seamless data integration.

Technical Associate *Aug 2021 – Oct 2023*
Inforge Technologies Pvt Ltd, Coimbatore, India

- Designed and developed scalable backend APIs using Node.js and Express.js.
- Improved data security by implementing secure authentication mechanisms, increasing protection by 40%.
- Reduced application crashes by 25% through debugging and performance optimization.

Education

MSc Artificial Intelligence *Jan 2025 – Jan 2026 (Expected)*
Aston University, Birmingham, UK

BSc Computer Science *Jun 2017 – May 2020*
Bharathiar University, Erode, India

Hobbies and Interests

Exploring AI innovations and applications, Learning new programming languages, Playing chess